HORIZON-WIDERA-2023-ERA-01-11: Research ethics for environmental and climate technologies

Specific conditions	
Expected EU contribution per project	The Commission estimates that an EU contribution of around EUR 3.00 million would allow these outcomes to be addressed appropriately. Nonetheless, this does not preclude submission and selection of a proposal requesting different amounts.
Indicative budget	The total indicative budget for the topic is EUR 3.00 million.
Type of Action	Coordination and Support Actions
Eligibility	The conditions are described in General Annex B. The following

[76] Shemakes.EU and EQUALS-EU

[77] https://cordis.europa.eu/project/id/665566; https://cordis.europa.eu/project/id/665825; https://cordis.europa.eu/project/id/101000063

conditions	exceptions apply: Due to the scope of this topic, legal entities established in non-associated third countries are exceptionally eligible for Union funding. Moreover, due to the scope of this topic, applications must be submitted by consortium and the minimum eligibility criteria for consortium composition set out in part B of the General Annexes applies. In addition, proposals must include at least two additional participants (joining as an associated partner or beneficiary if eligible for funding), each of whom must be a legal entity established in one of the following countries: Japan, China, Republic of Korea, or African countries not-associated to Horizon Europe.
Legal and financial set-up of the Grant Agreements	The rules are described in General Annex G. The following exceptions apply: Beneficiaries will be subject to the following additional dissemination obligations: Proposals must include structured cooperation (including the necessary technical aspects) with the e-platform Embassy of Good Science[78]. The output material of the action must be made available on this e-platform.

Expected Outcome: In order to promote a responsible implementation of the EU Green deal, the projects are expected to contribute to the following outcomes:

- Design an operational ethics and integrity framework, which preserves and promotes the key ethics principles while supporting a rapid and effective green transition in the European Union;
- Promote awareness, ethics education and training about climate and environmental aspects of research activities, as well as insight in ethical aspects of the development of related knowledge and applications (for example: new agricultural and breeding techniques, environmental protection, geoengineering, tools facilitating energy efficiency and behavioural change).

Scope: Becoming the world's first climate-neutral continent by 2050 is the greatest challenge and opportunity of our times. For this reason, the European Commission adopted the European Green Deal, the most ambitious package of measures that should enable European citizens and businesses to benefit from a sustainable green transition. In order to support the green transition, it is a priority for the European Research Area to build an encompassing framework for research and innovation activities[79].

78 www.embassy.science
79 Council Conclusions on Biodiversity – the need for urgent action, 11829/20, https://data.consilium.europa.eu/doc/document/ST-11829-2020-INIT/en/pdf

The high magnitude and multi-fold nature of the consequences that we would face, if we do not tackle the global environmental risks, necessitate adapting the way we work, protect the world's scarce resources, and decide on policy priorities. This inevitably raises important ethical questions and dilemmas including some related to the production of scientific knowledge and the development of novel technologies.

There is a growing awareness that biodiversity loss and ecosystem degradation at local, regional and global scale pose direct and existential threats to human life and wellbeing[80]. At the same time, actively pursuing the preservation of the environment can, in some cases, lead to some tensions between the pure environmental objective and the protection of human rights. This is the case in particular when the implementation of anthropomorphic models gives to natural elements the same status as human beings.

An important aspect that characterises global challenges is that they, by nature, go beyond the well-being of persons and touch the whole society notably in terms of solidarity and social justice. Mid and long-term socio-economic consequences are also more prominent in these complex research contexts, not only those affecting vulnerable populations exposed to environmental degradation, but also those caused by a green transition depending on social and geographic circumstances[81].

Some research topics intrinsically also have a complex environmental and ecological ethics dimension including for example research and innovation in the area of electro-magnetic fields and the high frequency communication systems that are necessary to achieve a European gigabit society, or the digital innovation and biotechnology in food production practices that should not only remain safe for human health and the environment, but also allow a fair and sustainable system[82].

In this context, the action should conduct an analytical work covering the following aspects:

1. What characterises the different dimensions and concepts associated with climate and environmental ethics in the context of research and development;
2. Identify the ethics and integrity challenges related to the production and use of scientific knowledge in designing and implementing novel technologies and approaches to the global environmental challenge facing the European Union and the Planet;
3. Develop strategies to uphold the integrity of scientific research in addressing climate change issues[83].

Elements to tackle by this action should encompass the issues related to the role of ethics and integrity experts (as advisors, for example), informed consent of communities and

80 Council Conclusions on Biodiversity – the need for urgent action, 11829/20, https://data.consilium.europa.eu/doc/document/ST-11829-2020-INIT/en/pdf
81 https://ec.europa.eu/info/sites/info/files/european-green-deal-communication_en.pdf
82 Council Conclusions on the Farm to Fork Strategy, 12099/20, https://data.consilium.europa.eu/doc/document/ST-12099-2020-INIT/en/pdf
83 UNESCO Declaration of Ethical Principles in Relation to Climate Change (2017), https://en.unesco.org/themes/ethics-science-and-technology/ethical-principles.

individuals[84], undue inducement and opt out approach, as well as equitable sharing of benefits arising from research[85].

The action should clearly highlight what cannot be accepted or neglected in the name of addressing environmental issues. This notably includes the need to always conduct, prior to the start of a research, an independent ethical review, which remains a necessary safeguard for the individuals involved and enhances the trust from the impacted communities and the society as a whole, in the name of the 'do no harm' principle[86]. Environmental concerns justify immediate actions and should not lower ethics and integrity standards.

In addition, issues related to refining environmental risk assessment in various fields of research and innovation should be addressed. The action should explore also how the quality of data estimating environmental impact is assessed and fed back in policy design. This action supports ERA Policy Agenda actions 11 and 12.

The action should result in:

- Producing an operational ("how-to") guidelines to support the work of research teams' ethics committee members and integrity experts, taking into account the concept of climate justice, including intergenerational justice as well as gender justice. The guidelines should include, among others, clear guidance for addressing ethical challenges related to the development of novel technologies and approaches to address climate change (e.g., in relation to technologies encouraging behavioural change, geo-engineering) and the application of the precautionary approach in different fields of research and innovation;

- Assessing the need to complement the European Code of Conduct for Research Integrity [87] with specific guidelines and if relevant propose short documents complementing the Code, focusing on the need to ensure an "inclusive and just transition that leaves no one behind"[88];

- An effective incorporation of the objectives of the "Do No Significant Harm" Principle[89];

- Developing traditional and online training material (reflecting the guidelines) for students, early career and experienced researchers. The material must be made available

84 As also highlighted in the Council Conclusions on Biodiversity – the need for urgent action, 11829/20, https://data.consilium.europa.eu/doc/document/ST-11829-2020-INIT/en/pdf

85 Ibid.

86 Also reaffirmed in the Council Conclusions on Biodiversity – the need for urgent action, 11829/20, https://data.consilium.europa.eu/doc/document/ST-11829-2020-INIT/en/pdf

87 http://ec.europa.eu/research/participants/data/ref/h2020/other/hi/h2020-ethics_code-of-conduct_en.pdf

88 https://ec.europa.eu/info/sites/info/files/european-green-deal-communication_en.pdf

89 https://ec.europa.eu/info/funding-tenders/opportunities/docs/2021-2027/horizon/guidance/programme-guide_horizon_en.pdf

on the e-platform Embassy of Good Science[90]. The priorities of the European Digital Education Plan must be taken into account;

- The action should in this context foresee the training of 400-450 Horizon Europe ethics appraisal scheme experts, paying close attention to gender balance, as well as to gender equality and diversity related ethical aspects, and make use of their feedback to improve the trainings.

Overall, the work should be based on existing know how and have a bottom-up approach, involving all relevant stakeholders (e.g., researchers, research funders, policy-makers, publishers, citizens, civil society organisations) through the organisation of participatory events (workshops, consultations, 'town hall" meetings). Every effort should be made to achieve a 45% - or higher- female participation, especially among students, researchers, and ethics experts.

- The activities should propose ways and means to encourage changes in the research culture and promote openness, communication, dialogue and stronger links among stakeholders. This work should involve relevant ethics and integrity networks, such as ENRIO[91] or European networks of (early) career researchers and educators in the field of research ethics and integrity.

In order to improve the impact of the expected output (such as effectiveness of training courses, guidelines, toolboxes, etc.), cooperation with research management offices and ethics officers in Research Performing Organisations is highly recommended. In addition, National Contact Points should be provided with all the materials relevant to support their advisory activities.

Proposals should ensure that the publicly available results from relevant EU funded research projects (e.g., SOP4RI, Integrity, TRUST, Path2Integirty, TechEthos) [92] are taken into account. Budgeted cooperation (including the necessary technical aspects) with Embassy of Good Science[93] should be included.

In order to achieve the expected outcomes, cooperation with at least two participants from Japan, China, the Republic of Korea and/or African countries non-associated to Horizon Europe is required.

Consortia with EU partners or partners from Associated Countries that have not previously collaborated are encouraged to participate.

For all published articles and deliverables produced in the context of the activities, an authorship contribution statement must be added, in accordance with a recognised standardised taxonomy developed for this purpose (e.g., CRediT).

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90 www.embassy.science
91 http://www.enrio.eu/
92 Detailed information of the mentioned EU funded projects can be found on CORDIS website: https://cordis.europa.eu/
93 www.embassy.science